

Problem 1: Sanskrit Months (15 marks)

The traditional Indian calendars, collectively known as the Hindu calendars, follow a luni-sidereal system of time keeping, where the movement of moon against the fixed background of stars is used as the basis for defining years, months and dates. While the Hindu calendars followed in various regions of India today vary greatly from each other, one thing that is common to all of them is the names of the 12 months. The Sanskrit names for these months, in the temporal order, are:

Caitra, Vaiśākha, Jyēṣṭha, Āṣāḍha, Śrāvaṇa, Bhādrapada, Āśvina, Kārtika, Mārgaśīrṣa, Pauṣa, Māgha, Phālguna

The month of *Caitra* coincides with mid-March or mid-April in most of the modern Hindu calendars. Interestingly, these names have been systematically derived from the names of the *nakṣatra*'s (stars or constellations) that coincide with the moon in that month on the full-moon night (*Pūrṇimā*). There are 27 *nakṣatra*'s corresponding to the 27 days of the lunar cycle.

Given below are the names of 23 of the 27 *nakṣatra*'s, in the right order.

Aśvinī, Bharanī, Kṛttikā, Rohiṇī, Mṛgaśīrṣa, Ārdrā, Punarvasu, Puṣya, Aślēṣā, Pūrva Phalgunī, Uttara Phalgunī, Hastā, Svāti, Viśākhā, Anurādhā, Jyēṣṭha, Mūla, Śrāvaṇa, Śraviṣṭhā, Śatabhiṣak, Pūrva Bhādrapadā, Uttara Bhādrapadā, Revatī

It is possible to guess the names and positions of the four missing *nakṣatra*'s if you use the following information:

1. All of them end with *ā*.
2. All of them begin with a short vowel. (Not really, but you can assume so.)
3. There is no missing *nakṣatra* between *Jyēṣṭha* and *Mūla*

Assignment 1:

Assuming *Aśvinī* to be the first *nakṣatra* (and *Revatī* the 27th), write down the positional numbers and the names of the 4 missing *nakṣatra*'s in the list.

Assignment 2:

Write down the names of the 12 *nakṣatra*'s with whom the full moon coincides for each of the 12 months.

Assignment 3:

The following 4 *nakṣatra*'s do not lend their names to any month:

Hastā, Punarvasu, Śatabhiṣak, Revatī

However, if one had to derive names of months from them, what would they be? In case it is not possible to predict the month name for some of these *nakṣatra*'s, write "NP" in the Answer Booklet.

Explain your reasoning, and then make your best guess for what the month name might be.

1. *h* indicates that the preceding consonant is aspirated (pronounced with a puff of air); *ṇ*, *ṣ*, *ṭ* denotes *n*, *sh* and *t* in *barn*, *marsh* and *art*, uttered with the tip of the tongue turned back (also colloquially referred to as the hard consonants); *ś* is pronounced as *sh* as in *shoe* and *s* is pronounced as *s* in *sea*; *r* is a vowel; a bar over a letter (e.g., *ā*, *ī*) denotes a long vowel. **2.** *Sanskrit* is the primary sacred language of Hinduism, a philosophical language in Buddhism, Hinduism and Jainism, and a literary language that was in use as a lingua franca in Greater India. It is a standardized dialect of *Old Indo-Aryan*, originating as *Vedic Sanskrit* and tracing its linguistic ancestry back to *Proto-Indo-Iranian* and *Proto-Indo-European*. Today, it is listed as one of the 22 scheduled languages of India. **3.** The lunar cycle is of 27 days and 7¾ hours; the 27 days are covered by the 27 *nakṣatra*'s and the fractional part is compensated by an intercalary 28th *nakṣatra* titled *Abhijit*. *Nakṣatra* computation appears to have been well known at the time of the *R̥gveda*.